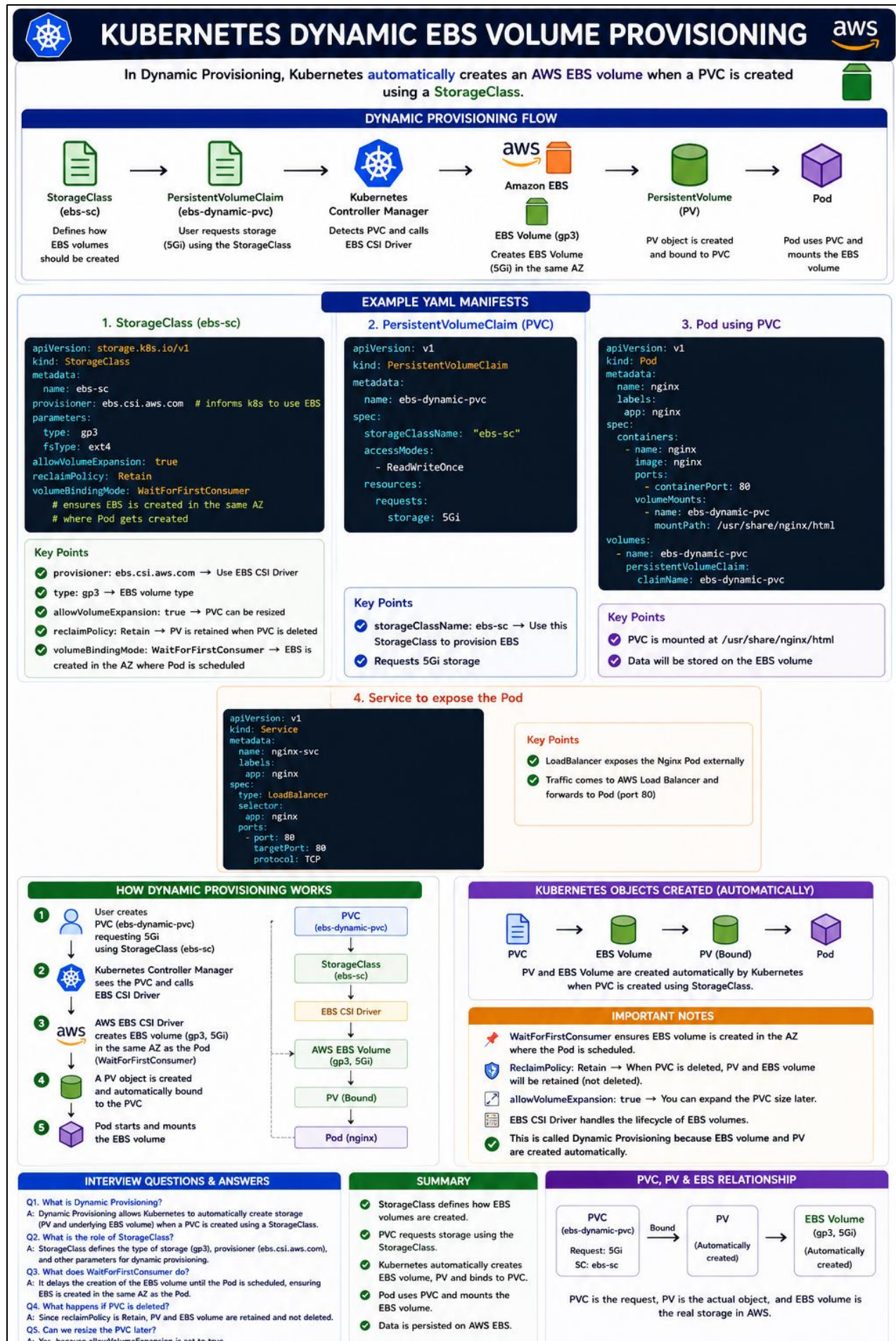


## Kubernetes Dynamic EBS Volume provisioning using StorageClass



## Dynamic Provisioning on EKS using Amazon EBS (with CSI Driver)

Dynamic provisioning allows Kubernetes to automatically create and manage EBS volumes when a pod requests storage. Unlike static provisioning, you do not manually create PVs; Kubernetes handles volume creation using a StorageClass and the EBS CSI driver.

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
```

```
$ eksctl create cluster --name dyanebs --nodegroup-name b17dng --node-type m7i-flex.large --nodes 3 --managed
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
```

```
$ eksctl get cluster
```

| NAME    | REGION    | EKSCTL CREATED |
|---------|-----------|----------------|
| dyanebs | us-east-1 | True           |

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
```

```
$ kubectl get nodes
```

| NAME                           | STATUS | ROLES  | AGE   | VERSION             |
|--------------------------------|--------|--------|-------|---------------------|
| ip-192-168-16-95.ec2.internal  | Ready  | <none> | 3m43s | v1.34.9-eks-93b80c6 |
| ip-192-168-41-189.ec2.internal | Ready  | <none> | 3m41s | v1.34.9-eks-93b80c6 |
| ip-192-168-8-162.ec2.internal  | Ready  | <none> | 3m43s | v1.34.9-eks-93b80c6 |

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes (main)
```

```
$ kubectl apply -k "github.com/kubernetes-sigs/aws-ebs-csi-driver/deploy/kubernetes/overlays/stable/?ref=release-1.51"
```

serviceaccount/ebs-csi-controller-sa created

serviceaccount/ebs-csi-node-sa created

role.rbac.authorization.k8s.io/ebs-csi-leases-role created

clusterrole.rbac.authorization.k8s.io/ebs-csi-node-role created

clusterrole.rbac.authorization.k8s.io/ebs-external-attacher-role created

clusterrole.rbac.authorization.k8s.io/ebs-external-provisioner-role created

clusterrole.rbac.authorization.k8s.io/ebs-external-resizer-role created

clusterrole.rbac.authorization.k8s.io/ebs-external-snapshotter-role created

rolebinding.rbac.authorization.k8s.io/ebs-csi-leases-rolebinding created

clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-attacher-binding created

clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-node-getter-binding created

clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-provisioner-binding created

clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-resizer-binding created

clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-snapshotter-binding created

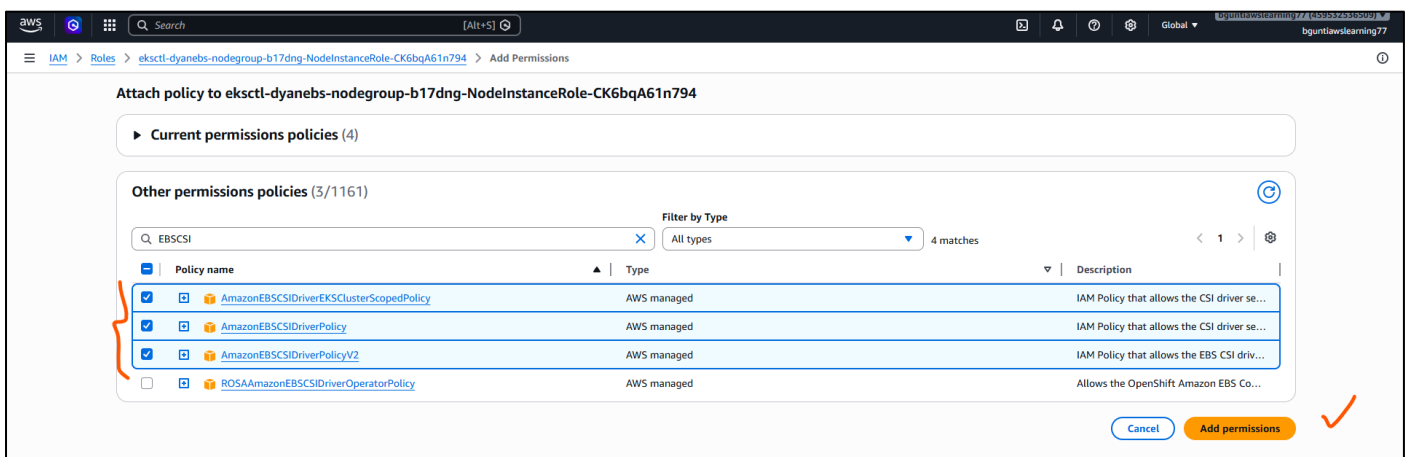
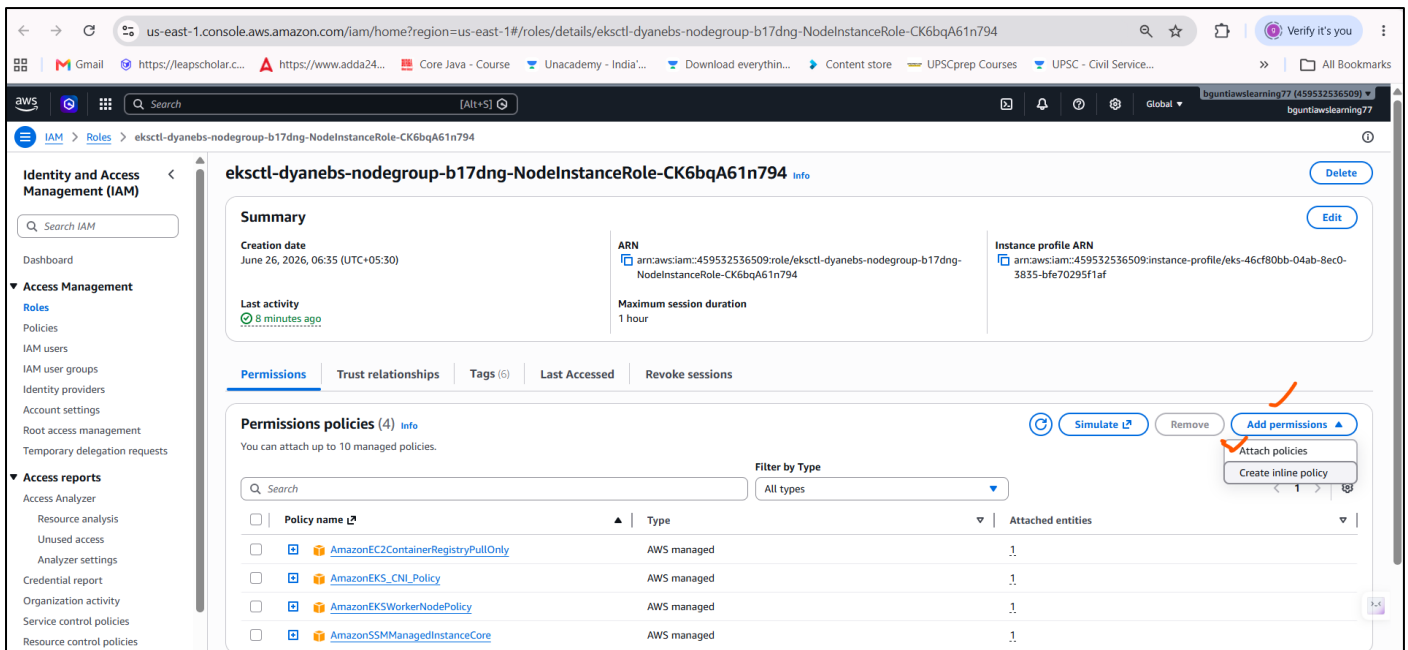
deployment.apps/ebs-csi-controller created

poddisruptionbudget.policy/ebs-csi-controller created

daemonset.apps/ebs-csi-node created

csidriver.storage.k8s.io/ebs.csi.aws.com created

The screenshot displays the AWS Management Console interface for an EKS cluster named 'dyanebs'. The 'Instances' tab is active, showing a list of EC2 instances. The instances are named 'dyanebs-b17dng-Node' and are in the 'Running' state. The 'Security' tab is also visible, showing the IAM role 'eksctl-dyanebs-nodegroup-b17dng-NodeInstanceRole' and the security groups associated with the instances. The console shows details for the instance 'i-04d9a55e418ccedd' (dyanebs-b17dng-Node), including its IAM role, security groups, and inbound rules.



Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops\_MC\_B17D/Kubernetes (main)

```
$ cd practice/25_June-K8s-Dynamic-EBS/
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops\_MC\_B17D/Kubernetes/practice/25\_June-K8s-Dynamic-EBS (main)

```
$ ll
```

```
total 1793
```

```
-rw-r--r-- 1 Bhavana 197121    568 Jun 26 06:36 1-storageclass-pvc.yaml
-rw-r--r-- 1 Bhavana 197121    632 Jun 26 06:38 2-pod-dynamic-ebs-svc.yaml
-rw-r--r-- 1 Bhavana 197121 1828149 Jun 26 06:35 dynamic-ebs.png
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops\_MC\_B17D/Kubernetes/practice/25\_June-K8s-Dynamic-EBS (main)

```
$ kubectl apply -f 1-storageclass-pvc.yaml
storageclass.storage.k8s.io/ebs-sc created
persistentvolumeclaim/ebs-dynamic-pvc created
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops\_MC\_B17D/Kubernetes/practice/25\_June-K8s-Dynamic-EBS (main)

```
$ kubectl get sc
NAME      PROVISIONER      RECLAIMPOLICY  VOLUMEBINDINGMODE
ALLOWVOLUMEEXPANSION  AGE
ebs-sc    ebs.csi.aws.com  Retain         WaitForFirstConsumer  true
27s
gp2       kubernetes.io/aws-ebs  Delete        WaitForFirstConsumer  false
23m
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops\_MC\_B17D/Kubernetes/practice/25\_June-K8s-Dynamic-EBS (main)

\$ kubectl get pvc

| NAME            | STATUS  | VOLUME | CAPACITY | ACCESS MODES | STORAGECLASS |
|-----------------|---------|--------|----------|--------------|--------------|
| ebs-dynamic-pvc | Pending |        |          |              | ebs-sc       |
| <unset>         | 53s     |        |          |              |              |

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops\_MC\_B17D/Kubernetes/practice/25\_June-K8s-Dynamic-EBS (main)

\$ kubectl apply -f 2-pod-dynamic-ebs-svc.yaml

pod/nginx-pod-1 created  
service/nginx-svc created

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops\_MC\_B17D/Kubernetes/practice/25\_June-K8s-Dynamic-EBS (main)

\$ kubectl get all

| NAME            | READY | STATUS            | RESTARTS | AGE |
|-----------------|-------|-------------------|----------|-----|
| pod/nginx-pod-1 | 0/1   | ContainerCreating | 0        | 9s  |

| NAME               | TYPE         | CLUSTER-IP    | EXTERNAL-IP                                                            |
|--------------------|--------------|---------------|------------------------------------------------------------------------|
| service/kubernetes | ClusterIP    | 10.100.0.1    | <none>                                                                 |
| service/nginx-svc  | LoadBalancer | 10.100.140.48 | a5148e606b44a480d9746e9c2f7d6307-156314100.us-east-1.elb.amazonaws.com |
| 443/TCP            |              |               | 80:30479/TCP                                                           |

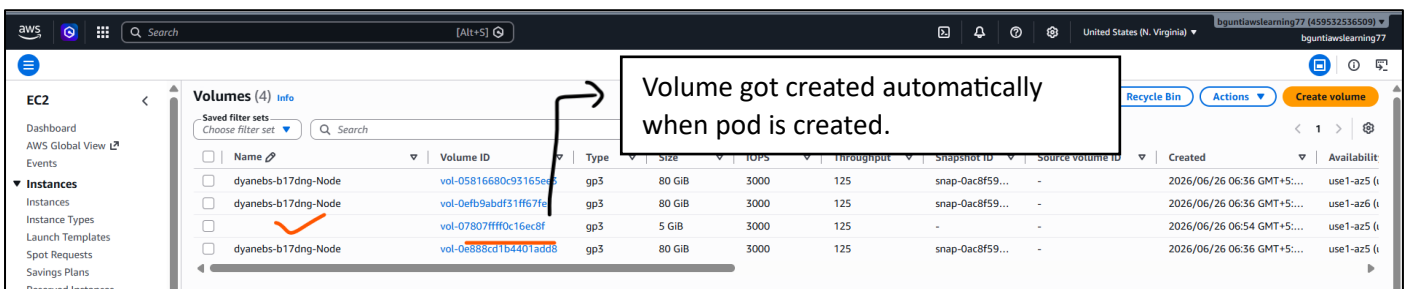
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops\_MC\_B17D/Kubernetes/practice/25\_June-K8s-Dynamic-EBS (main)

\$ kubectl get all

| NAME            | READY | STATUS  | RESTARTS | AGE |
|-----------------|-------|---------|----------|-----|
| pod/nginx-pod-1 | 1/1   | Running | 0        | 22s |

| NAME               | TYPE         | CLUSTER-IP    | EXTERNAL-IP                                                            |
|--------------------|--------------|---------------|------------------------------------------------------------------------|
| service/kubernetes | ClusterIP    | 10.100.0.1    | <none>                                                                 |
| service/nginx-svc  | LoadBalancer | 10.100.140.48 | a5148e606b44a480d9746e9c2f7d6307-156314100.us-east-1.elb.amazonaws.com |
| 443/TCP            |              |               | 80:30479/TCP                                                           |



```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/25_June-K8s-Dynamic-EBS (main)
$ kubectl exec -it nginx-pod-1 -- bash
root@nginx-pod-1:/# ls
bin boot dev docker-entrypoint.d docker-entrypoint.sh etc home lib
lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@nginx-pod-1:/# cd /usr/share/nginx/html/
root@nginx-pod-1:/usr/share/nginx/html# ls
lost+found
root@nginx-pod-1:/usr/share/nginx/html# echo "<h1>Hello, This content is
shared from Dynamic EBS Volume created using StorageClass.</h1>" > index.html
root@nginx-pod-1:/usr/share/nginx/html# cat index.html
<h1>Hello, This content is shared from Dynamic EBS Volume created using
StorageClass.</h1>
root@nginx-pod-1:/usr/share/nginx/html# exit
exit

```



```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/25_June-K8s-Dynamic-EBS (main)
$ kubectl get pvc
NAME                STATUS    VOLUME                                     CAPACITY
ACCESS MODES        STORAGECLASS  VOLUMEATTRIBUTESCLASS  AGE
ebs-dynamic-pvc     Bound       pvc-19e5bdc5-0689-4feb-8c37-508aab05a467  5Gi
RWX                  ebs-sc       <unset>                  12m

```

```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/25_June-K8s-Dynamic-EBS (main)
$ kubectl delete po nginx-pod-1
pod "nginx-pod-1" deleted from default namespace

```

```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/25_June-K8s-Dynamic-EBS (main)
$ kubectl apply -f 2-pod-dynamic-ebs-svc.yaml
pod/nginx-pod-2 created
service/nginx-svc unchanged

```

```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/25_June-K8s-Dynamic-EBS (main)
$ kubectl get all
NAME                READY    STATUS    RESTARTS   AGE
pod/nginx-pod-2     1/1      Running   0           18s

NAME                TYPE          CLUSTER-IP      EXTERNAL-IP
PORT(S)            AGE           service/kubernetes  ClusterIP   10.100.0.1      <none>
443/TCP             36m
service/nginx-svc   LoadBalancer 10.100.140.48
a5148e606b44a480d9746e9c2f7d6307-156314100.us-east-1.elb.amazonaws.com
80:30479/TCP       12m

```

```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/25_June-K8s-Dynamic-EBS (main)
$ kubectl get pvc
NAME                STATUS    VOLUME                                     CAPACITY
ACCESS MODES        STORAGECLASS  VOLUMEATTRIBUTESCLASS  AGE
ebs-dynamic-pvc     Bound       pvc-19e5bdc5-0689-4feb-8c37-508aab05a467  5Gi
RWX                  ebs-sc       <unset>                  14m

```



```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/25_June-K8s-Dynamic-EBS (main)
```

```
$ eksctl get cluster
```

| NAME    | REGION    | EKSCTL CREATED |
|---------|-----------|----------------|
| dyanebs | us-east-1 | True           |

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/25_June-K8s-Dynamic-EBS (main)
```

```
$ eksctl delete cluster dyanebs
```